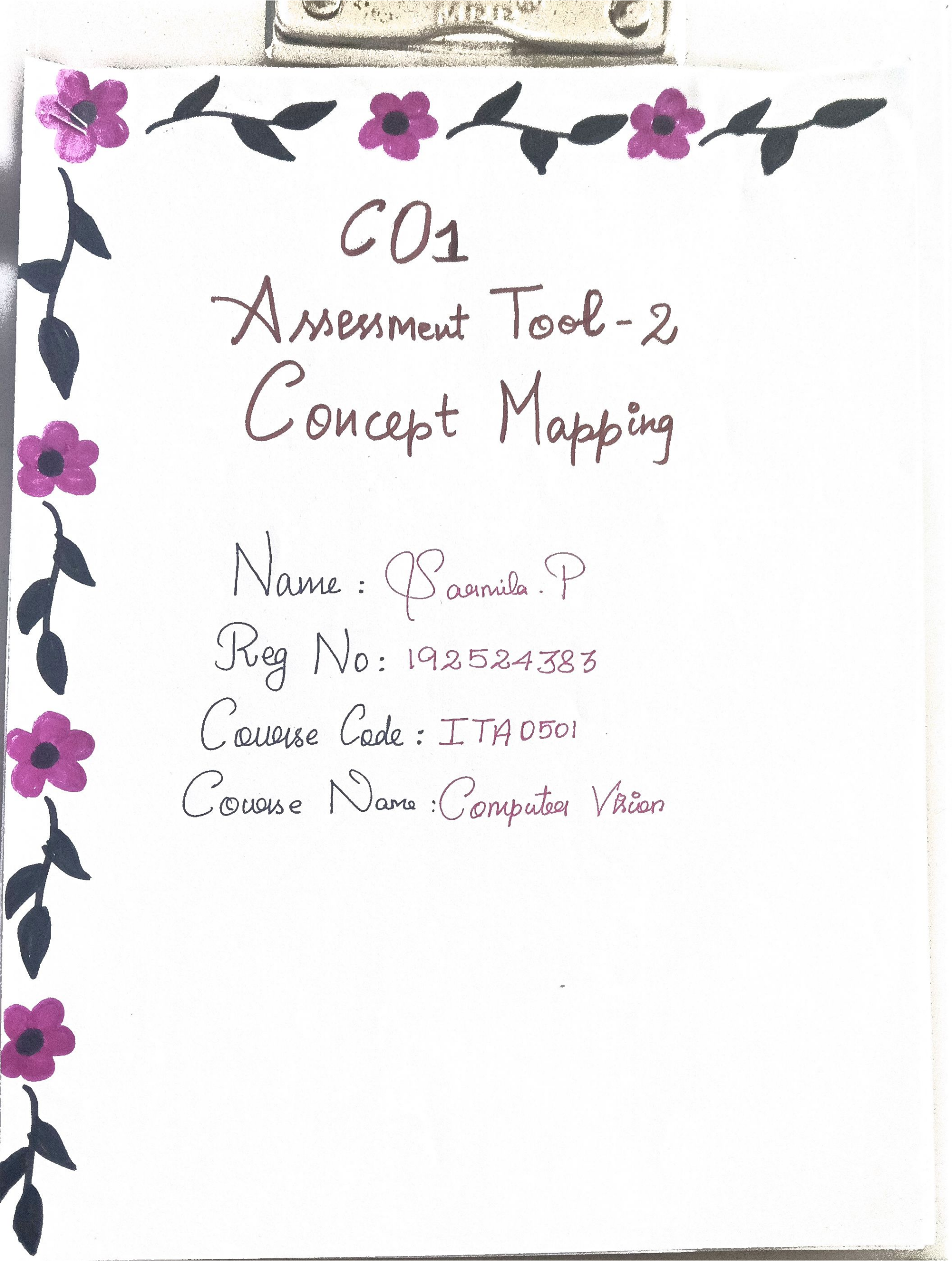




CO1

Assessment Tool-2
Concept Mapping

Name : Saimila . P

Reg No : 192524383

Course Code : ITA0501

Course Name : Computer Vision

Overview & Features:

- Helps machines understand visual information.
- Work with images/videos.

Feature:

- Detection
- Recognition
- Tracking



Computer Vision:



- Image Capture
- Image Processing
- Feature Extraction
- Object Recognition
- Navigation

Digital Image Fundamentals:

- Pixel
- Resolution
- Intensity
- Grayscale/Color



Image Sensing & Acquisition

Camera → Sensor → Digital Image

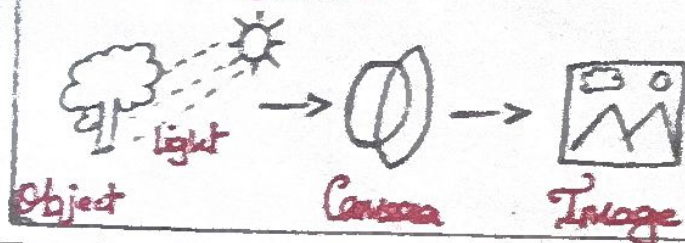
Capture Scene

Convert light to image



Image Formation Model:

- Object
- Illumination
- Lens
- Sensor



Autonomous Vehicle Computer Vision Based Navigation

Levels of Computer Vision

Low Level
Image Processing

Mid level
Feature Extraction

High level
Object Recognition

Sampling & Quantization:

Continuous Image → Digital Image

Sampling → Spatial coordinates

Quantization → Intensity values



Continuous



Digital

Application:

- Road Detection
- Lane Detection
- Navigation
- Traffic Sign Recognition



Digital Image

Fundamental:

- *) Pixel
- *) Resolution
- *) Intensity
- *) Grayscale/Color

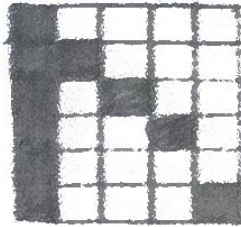


Image Sensing & Acquisition

- *) X-ray source
- *) Detector (digital sensor)
- *) Capture X-ray image
- *) Convert to digital signal

Sampling & Quantization:

*) Continuous image
→ Digital image

*) Sampling → Spatial coordinates



Continuous



Digital

Medical Imaging

Capture → Process → Diagnose

Application in Medical Imaging

Disease detection (eg. fracture)
Tumor / lesion identification
Organ analysis (lungs, heart)
Treatment planning



Dist. X-ray

CT Scan

Features of Computer Vision

- *) Detection
- *) Recognition
- *) Tracking
- *) Classification

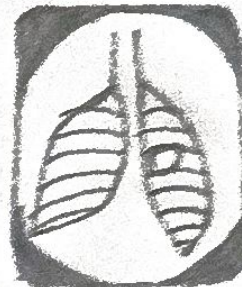


Image Formation Model:

- *) X-ray photons
- *) Patient
- *) Detector
- *) Image



X-ray source



Patient



Detector